MCP Server Kubernetes + Claude AI Integration (On-Prem Setup)

Overview:

This setup integrates:

- mcp-server-kubernetes
- Claude
- Kubernetes cluster (kubeadm – 4 nodes)

Purpose:
Control Kubernetes using **natural language via AI**

Architecture:

```
Claude CLI (AI Agent)
      ↓ (stdio)
MCP Server (Node.js)
      ↓
Kubernetes Cluster (via kubeconfig)
```

Prerequisites

- Ubuntu 20.04 / 22.04 server
- Kubernetes cluster (kubeadm)
- kubectl configured
- Internet access

Step 1: Clone Repository

```
sudo su
cd ~
git clone https://github.com/Flux159/mcp-server-kubernetes.git
cd mcp-server-kubernetes
```

Step 2: Install Bun

```
curl -fsSL https://bun.sh/install | bash
source ~/.bashrc
bun –version
```

Install Node: above 20.X.X version

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
apt install nodejs -y
```

Step 3: Install Dependencies

```
bun install
```

If slow:

```
export BUN_INSTALL_CACHE_DIR=/tmp/bun-cache
```

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Step 4: Build Project

bun run build

This generates: $ tsc && shx chmod +x dist/*.js

Step 5: Configure Kubernetes Access (if not)

```
mkdir -p ~/.kube
sudo cp /etc/kubernetes/admin.conf ~/.kube/config
sudo chown $USER:$USER ~/.kube/config
kubectl get nodes
```

Step 6: Install Claude CLI

```
curl -fsSL https://claude.ai/install.sh | bash
echo 'export PATH="$HOME/.local/bin:$PATH"' >> ~/.bashrc
source ~/.bashrc
claude –help
```

Step 7: Login (IMPORTANT – On-Prem)

Use API Key method (recommended)

Create API Key

Open in browser:

https://console.anthropic.com

Generate API key

Login via CLI

claude login OR claude

Select:

2. Anthropic Console account (API usage)

paste that API key here , you will get succuessfully login prompt

Step 8: Configure MCP Server (Do it on for clouad Desktop only)

claude config edit

```
{
  "mcpServers": {
    "kubernetes": {
      "command": "node",
      "args": ["/home/<your-user>/mcp-server-kubernetes/dist/index.js"],
      "env": {
        "KUBECONFIG": "/home/<your-user>/.kube/config"
      }
    }
  }
```

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}

Replace `<your-user>` accordingly

Step 9: Start Claude

claude

Step 10: Test Integration

List all Kubernetes pods

Show all nodes in my cluster

Example Prompt (Nginx Deployment):

Create a complete Kubernetes setup in my cluster with the following requirements:
1. Create a namespace called "prod"
2. Create a PersistentVolume (PV):
   - Use local storage
   - Path: /tmp/nginxlog
   - Capacity: 1Gi
   - Access mode: ReadWriteOnce
3. Create a PersistentVolumeClaim (PVC):
   - Namespace: prod
   - Request: 1Gi
   - Access mode: ReadWriteOnce
4. Create a ConfigMap:
   - Namespace: prod
   - Name: nginx-html
   - Add custom index.html content like "Welcome to Production Nginx"
5. Create an Nginx Deployment:
   - Namespace: prod
   - Use nginx:latest image
   - Mount PVC to /var/log/nginx
   - Mount ConfigMap to /usr/share/nginx/html
6. Expose the deployment using a NodePort service Add Ssl for webiste create ssl direcrtoy in same folder..
7. Ensure all resources are properly linked and working..
After creating, verify all resources and show me:
- pods – pv – pvc - services

procced with yes only.

Verify Resources
kubectl get all -n prod
kubectl get pv
kubectl get pvc -n prod

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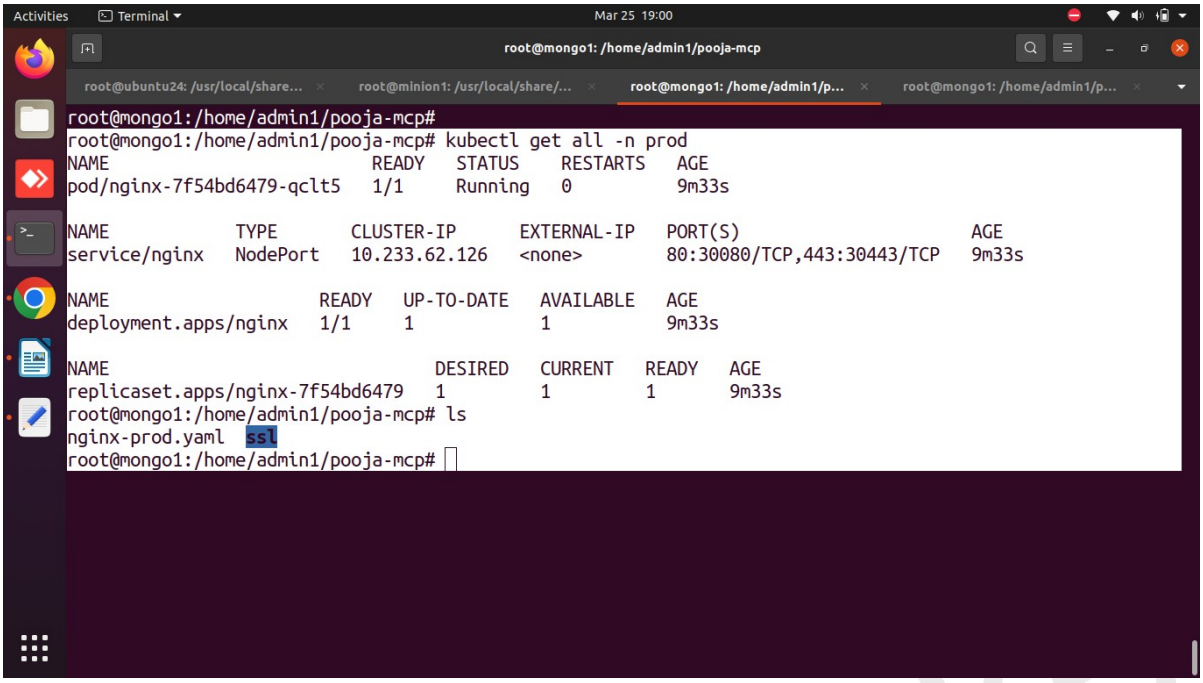
Activities Terminal Mar 25 18:54

✳ Delete production namespace setup

root@ubuntu24: /usr/local/share... root@minion1: /usr/local/share/... ✳ Delete production namespace ... root@mongo1: /home/admin1/p...

```
  kubectl get pods -n prod -o wide && \…)
═══ ═══ ═══ 
```



Security Best Practice

Avoid using admin.conf in production.

Create limited access:

kubectl create serviceaccount mcp-user

kubectl create clusterrolebinding mcp-binding \

--clusterrole=view \

--serviceaccount=default:mcp-user

Production Improvements

- Use Nginx as reverse proxy
- Add HTTPS (Let's Encrypt)
- Use restricted RBAC
- Monitor via Prometheus/Grafana

Key Concepts

Component	Role
Claude	AI agent
MCP Server	Middleware
Kubernetes	Execution layer

Permission error:

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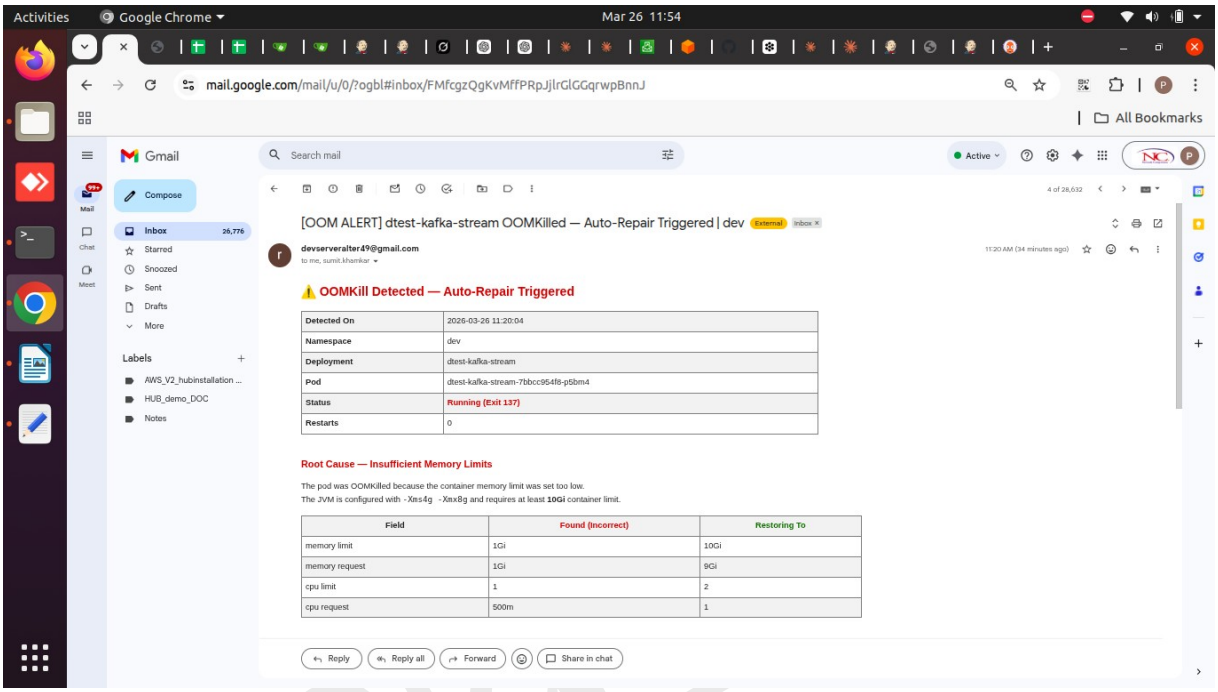
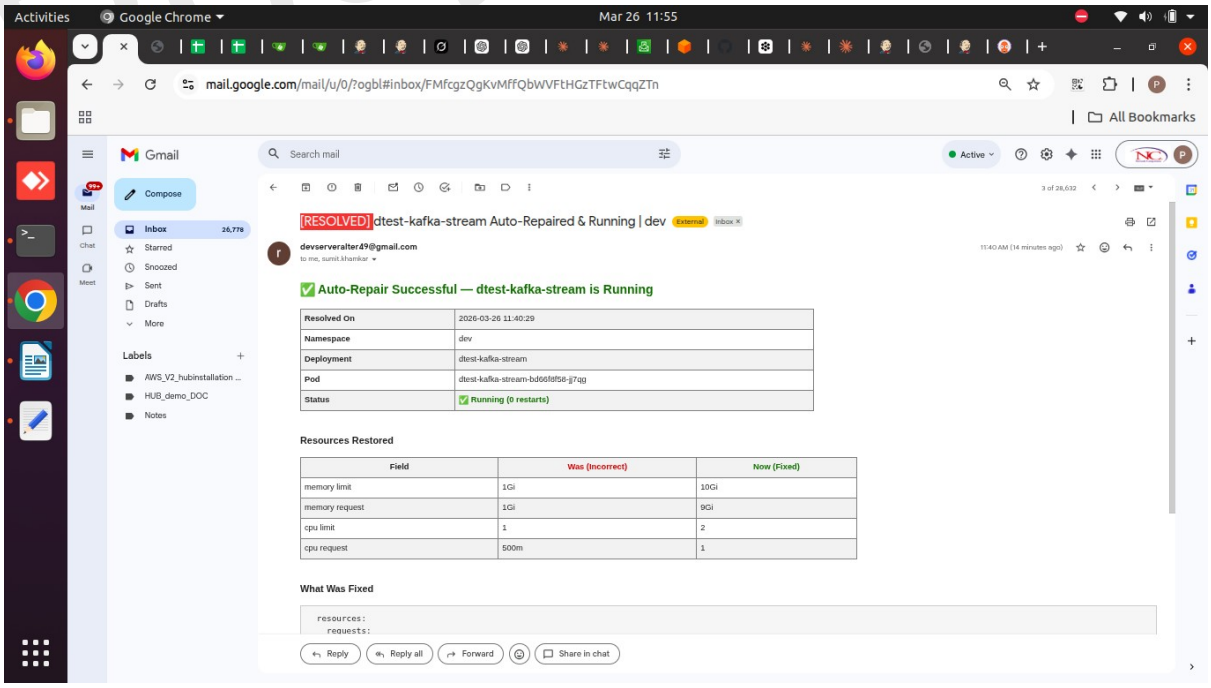
sudo chown -R $USER:$USER ~/mcp-server-kubernetes

Using Claude AI, I created an automated script that monitors the entire Kubernetes cluster, detects pod-level errors, and triggers email notifications. The system not only alerts on issues but also attempts automated fixes and sends a follow-up email once the issue is resolved.